



3DSS ASV Integration Supplementary

Revision 1.1 – 5 June 2023

This document provides some supplementary information and general guidelines for a 3DSS iDX integration on an ASV platform.

Recommended References:

3DSS Sonar Manual / Quick Start Guide

<https://ln5.sync.com/dl/aed4854b0/e8v6jhcm-s8mgtyf8-uvfnrjht-cf325rtx>

<https://ln5.sync.com/dl/b7fdb9310/nsfzx8sf-u3v3v86i-kw8w24p7-5tdimd9r>

Mechanical Drawings

<https://ln5.sync.com/dl/6589c6760/9rturwb5-yca34xwt-cmy666ne-tkhbugc4> (iDX Sonar Head)

<https://ln5.sync.com/dl/35d0a7ef0/bfr5vrrk-maxxtnag-fidwshdq-q83cvmxd> (Flange Mount)

<https://ln5.sync.com/dl/5dfdc720/x9fvjt27-zivz9kze-2k4mcihc-rr6bwqp4> (Topside SIU)

<https://ln5.sync.com/dl/3ced6dbf0/ngseqxt9-t7dhxfhj-ij5vbqzx-bs35tjmt> (Bulkhead Adapter)

Integration Checklist:

- 1) Sonar head should be mounted below the keel of the boat to avoid occlusion from the hull and to limit multipath interference. Three mounting options are possible utilizing the 3DSS-DX Pole Adapter, the 3DSS-DX Flange Adapter, or Direct Mounting to the sonar housing.
- 2) Sonar should ideally be mounted forward of props by 1m or as far forward as is feasible.
- 3) Topside Sonar Interface Unit (SIU) should be mounted in a dry area on the vehicle. A bulkhead adapter for connecting the Sonar to the SIU mounted in a dry box can be provided.
- 4) Primary and secondary antennas should ideally be spaced a minimum of 1m apart, located as far as possible from other RF equipment, have a free view of the sky, and be mounted at least 15 cm above the vessel surface.
- 5) Lever arms between the IMU and both primary and secondary antennas should be accurately measured (ideally to within 1cm) and entered into the INS configuration. See the '3DSS-IDX Quick Start Guide' for more detail.
- 6) RTCM corrections fed into the AUX port on the SIU at 115200 baud for precise positioning.
- 7) Low impedance power source of 12-24VDC connected to the SIU. Low impedance as defined as high current rated source (120W min) as well as short length and/or high gauge wiring from power source to SIU.



3DSS ASV Integration Supplementary

Revision 1.1 – 5 June 2023

- 8) On-board Windows PC with 10th generation i5 CPU or better (older CPU's with similar benchmarks can also be used), 16GB RAM, 512GB or greater SSD, and integrated gigabit Ethernet adapter.
- 9) Shore PC should include an integrated graphics adapter (NVIDIA GTX 1650 equivalent or better is recommended) for running the 3DSS 3D Display software.
- 10) Firewalls should be disabled, and the IP address of the network adapter used to communicate with the sonar should be configured to be on the 192.168.228 subnet. If multiple network adapters are used on the PC, the Interface Metric of the network adapter for the sonar should be set to '5', and other network adapter IP settings should not overlap with the 192.168.228 subnet.
- 11) The 3DSS software suite should be pre-installed and a valid license should be obtained either in the form of a hardware dongle or soft license key. During acquisition, it is recommended to record to disk the raw sonar data using the 3DSS Control software, as well as the raw GNSS / INS data using the 3DSS Binary Logger.

ASV Remote Sonar Control and Data Display:

A method of controlling the sonar, acquiring data, and viewing real time data remotely is required.

For this, a communication link between the ASV and shore PC is recommended which:

- Has a bandwidth of 2Mbps or greater
- Is setup such that both the vehicle PC and shore PC appear to be on the same LAN. This can be achieved by:
 - A point-to-point RF link (network bridge) between the vehicle and shore with connectivity via Ethernet (typical)
 - Internet connection (i.e. LTE) on both ends, with a VPN connection between them (advanced)

The typical solution consists of a Windows Remote Desktop (RDP) connection to the vehicle PC from the shore PC for sonar control, as well as the use of the 3DSS Compressor / Decompressor software to stream real-time sonar data back to the shore PC to be displayed.

The bandwidth requirements for Windows Remote Desktop (RDP) can vary depending on several factors, such as the screen resolution, color depth, and the complexity of the content being displayed or transmitted. Keep this mind when configuring the vehicle PC. For a typical remote desktop session, a rough estimate would be between 70-150 kilobits per second (Kbps) for lower resolutions and basic usage.



3DSS ASV Integration Supplementary

Revision 1.1 – 5 June 2023

The bandwidth requirement for the 3DSS Remote Data Link is between 2 Mbps and 4 Mbps depending on settings. The 3DSS Remote Data Link provides options for data decimation, and point rejection which can be used to adjust the Bandwidth requirements within that range if needed.

Software required on the vehicle PC :

- 3DSS-DX Sonar Control
- 3DSS-iDX Binary Logger
- 3DSS Remote Data Link software running in Compression Mode

Software required on the shore PC:

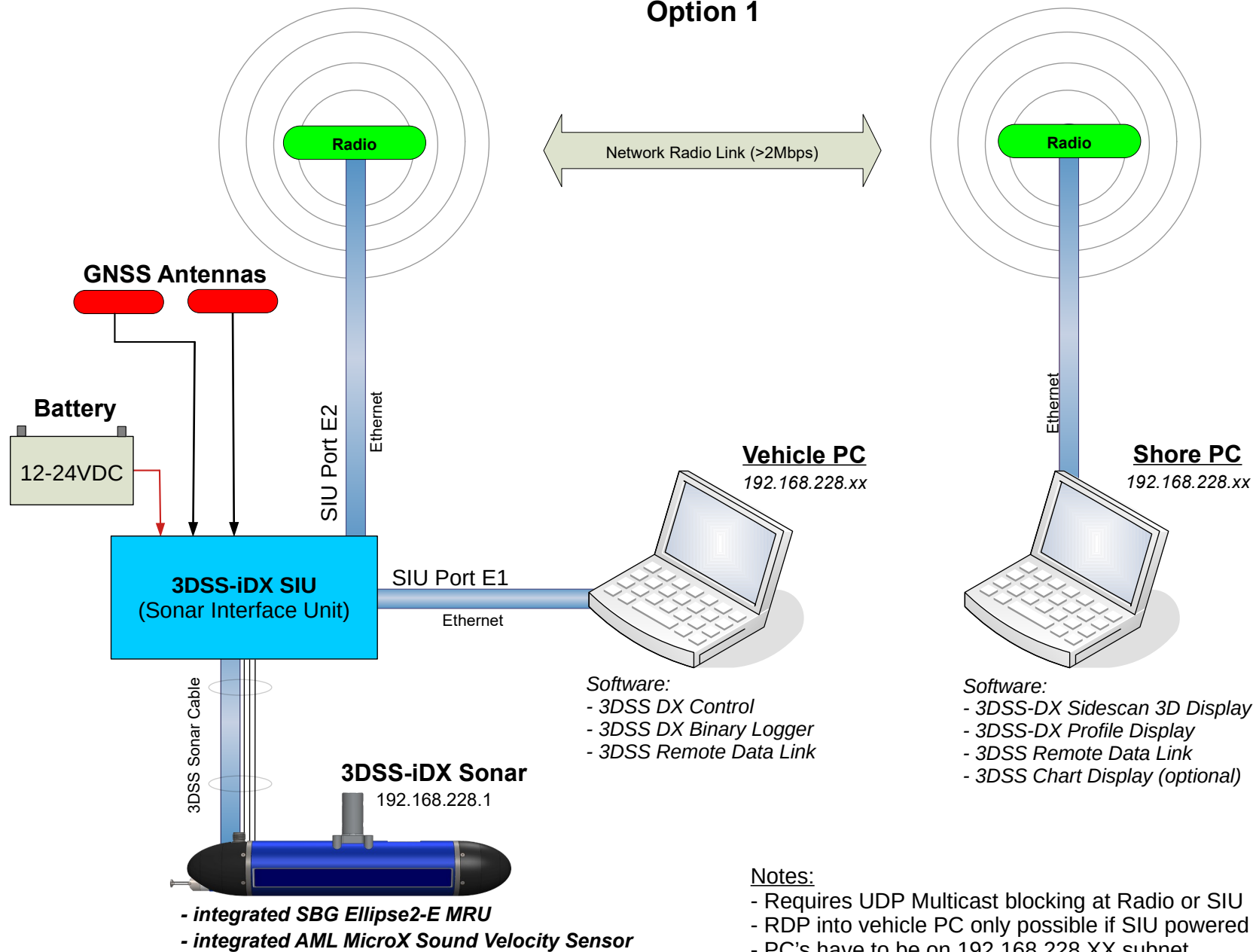
- 3DSS-DX Sidescan 3D Display
- 3DSS-DX Profile Display
- 3DSS Remote Data Link software running in Decompression Mode
- 3DSS Chart Display (optional)

Blocking Multi-Cast Packets from the radio link

The raw data from the sonar is transmitted via UDP using multicast. Care must be taken to prevent these packets from being sent over the radio as the volume of data can easily flood the bandwidth of a radio link. This can be done in a number of ways:

- a. If more than one Ethernet adapter is available on the vehicle PC, it is possible to keep the sonar and the radio on a separate network. See option 2 configuration below.
- b. By setting any of the below configurations on the radio or bridge device (if supported):
 - Disable IGMP: This prevents the devices from participating in multicast groups and effectively blocks multicast traffic.
 - Multicast Source Filtering: Multicast Source Filtering allows you to restrict multicast traffic based on the source IP address.
 - Access Control Lists (ACLs): Configure Access Control Lists on the bridge devices to explicitly deny or block multicast traffic. ACLs can be set up to filter based on source and destination IP addresses, protocol, or specific multicast group addresses.
- c. The Sonar Interface Unit (SIU) can be configured such that Ethernet port 'E2' blocks multicast packets while allowing them on 'E1'. In this configuration the SIU will act as an Ethernet switch where E2 is connected to the radio, and E1 is connected to the PC. The disadvantage to this setup is that if the SIU is not powered, the connection to the vehicle PC will be lost.

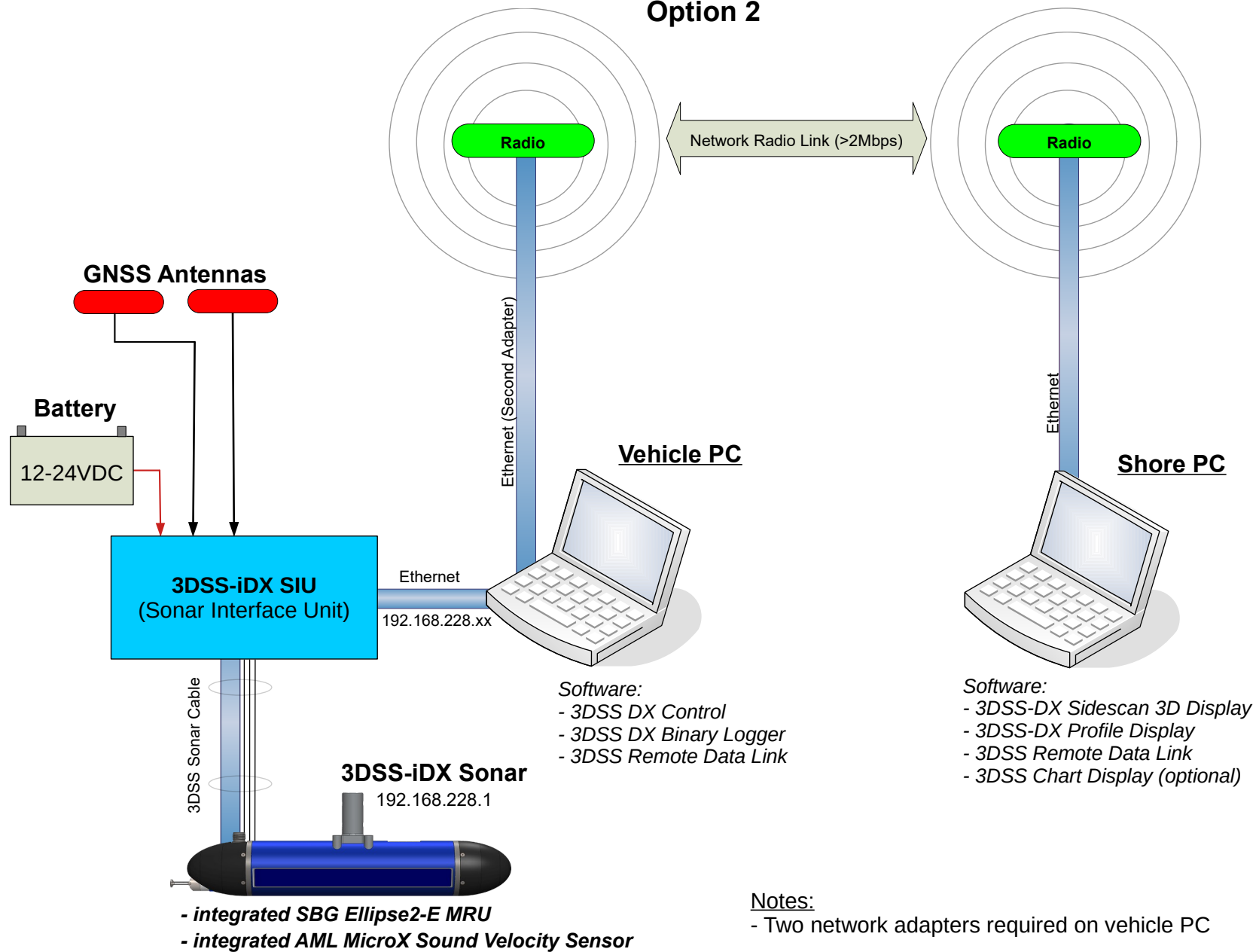
IDX USV Integration Diagram Option 1



Notes:

- Requires UDP Multicast blocking at Radio or SIU
- RDP into vehicle PC only possible if SIU powered
- PC's have to be on 192.168.228.XX subnet
- Only one network adapter required on vehicle PC

IDX USV Integration Diagram Option 2



ASV Data Flow

